

Lab Report No 8 AND 9
Z-TRANSFORM
Digital Signal Processing



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Lab Example No 1:

Solution:

Brief description (3-5 lines)

Poles, zeros and ROC of the expression of following form can determine using tf2zp command. Consider response of an LTI system.
 $G(z)=2z^4+16z^3+44z^2+56z+32/3z^4+3z^3-15z^2+18z-12$

The code

```
Z-TRANSFORM.PY > ...  
  Click here to ask Blackbox to help you code faster  
1  num=input('Type the Numerator coefficients=')  
2  import numpy as np  
3  from scipy import signal  
4  num=[2,16,44,56,32]  
5  den=[3,3,-15,18,-12]  
6  z,p,k=signal.tf2zp(num,den)  
7  roc=signal.convolve(den,den[::-1])  
8  print("Transfer function coefficients:")  
9  print("Numerator:",num)  
10 print("Denominator:",den)  
11 print("\nZeros:",z)  
12 print("Poles:",p)  
13 print("Gain",k)  
14 print("\nROC:",roc)
```

The results (Screenshot)

```
Type the Numerator coefficients=2,16,44,56,32  
Transfer function coefficients:  
Numerator: [2, 16, 44, 56, 32]  
Denominator: [3, 3, -15, 18, -12]  
  
Zeros: [-4.+0.j -2.+0.j -1.+1.j -1.-1.j]  
Poles: [-3.23606798+0.j 1.23606798+0.j 0.5 +0.8660254j  
0.5 -0.8660254j]  
Gain 0.6666666666666666  
  
ROC: [ -36  18 189 -522 711 -522 189 18 -36]
```

Lab Task No 1(i):

Solution:

Brief description (3-5 lines)

P-1) Consider the following difference equations:
Obtain the pole zero plot of the given systems, Analyze the above systems for stability, find their Fourier transform if possible. Compute the impulse response of the systems.

P-2) Find $h[n]$:

$$H_2(z) = \frac{2z^4 + 6.4z^3 + 4.9z^2 - 0.1z - 0.6}{5z^4 + 15.5z^3 + 13.7z^2 - 22.52z - 0.48}$$

The code

```
Z-TRANSFORM1.PY > ...  
  Click here to ask Blackbox to help you code faster  
1  num=input('Type the Numerator coefficients=')  
2  import numpy as np  
3  from scipy import signal  
4  num=[2,6.4,4.9,-0.1,-0.6]  
5  den=[5,15.5,13.7,-22.52,-0.48]  
6  z,p,k=signal.tf2zpk(num,den)  
7  roc=signal.convolve(den,den[::-1])  
8  print("Transfer function coefficients:")  
9  print("Numerator:",num)  
10 print("Denominator:",den)  
11 print("\nZeros:",z)  
12 print("Poles:",p)  
13 print("Gain",k)  
14 print("\nROC:",roc)
```

The results (Screenshot)

```

Type the Numerator coefficients=2,6.4,4.9,-0.1,-0.6
Transfer function coefficients:
Numerator: [2, 6.4, 4.9, -0.1, -0.6]
Denominator: [5, 15.5, 13.7, -22.52, -0.48]

Zeros: [-2.   -1.   -0.5   0.3]
Poles: [-1.93618735+1.41379112j -1.93618735-1.41379112j  0.79342587+0.j
        -0.02105117+0.j          ]
Gain 0.4

ROC: [  -2.4    -120.04   -287.136    -7.8644   960.3208    -7.8644   -287.136
       -120.04    -2.4    ]

```

Lab Task No 1(ii):

Solution:

Brief description (3-5 lines)

P-1) Consider the following difference equations:
Obtain the pole zero plot of the given systems, Analyze the above systems for stability, find their Fourier transform if possible. Compute the impulse response of the systems.

P-2) Find $h[n]$:

$$H_1(z) = \frac{(z+0.4)(z-0.91)(z^2+0.3z+0.4)}{(z^2-0.6z+0.6)(z^2+3z+5)}$$

The code

🔗 Z-TRANSFORM2.PY > ...

💡 Click here to ask Blackbox to help you code faster

```
1 num=input('Type the Numerator coefficients=')
2 import numpy as np
3 from scipy import signal
4 num=[1,-0.21,-0.1144,-0.3582,-0.1456]
5 den=[1,2.4,5.6,-1.2,3]
6 z,p,k=signal.tf2zpk(num,den)
7 roc=signal.convolve(den,den[::-1])
8 print("Transfer function coefficients:")
9 print("Numerator:",num)
10 print("Denominator:",den)
11 print("\nZeros:",z)
12 print("Poles:",p)
13 print("Gain",k)
14 print("\nROC:",roc)
```

The results (Screenshot)

```
Type the Numerator coefficients=1,-0.21,-0.1144,-0.3582,-0.1456
Transfer function coefficients:
Numerator: [1, -0.21, -0.1144, -0.3582, -0.1456]
Denominator: [1, 2.4, 5.6, -1.2, 3]

Zeros: [ 0.92930115+0.j          -0.1752308 +0.62775589j -0.1752308 -0.62775589j
        -0.36883955+0.j          ]
Poles: [-1.40319936+2.07181601j -1.40319936-2.07181601j  0.20319936+0.66169242j
        0.20319936-0.66169242j]
Gain 1.0

ROC: [ 3.    6.   19.52  5.52 48.56  5.52 19.52  6.    3.   ]
```

Conclusion

In conclusion, the Z-transform is a valuable tool for analyzing and designing discrete-time systems, providing insights into their behavior and enabling efficient signal processing techniques. Its applications extend across various domains, making it an essential concept in the field of engineering and mathematics.